Predicting Organized Opposition to Housing Development: Evidence from Austin, Texas

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Abstract

Can organized neighborhood opposition to housing development proposals be predicted before projects enter formal public hearings? Bridging a citywide longitudinal parcel footprint for development forecasting with an exhaustive target universe of 7,074 discretionary zoning events for opposition modeling, this research constructs a sequential predictive pipeline that forecasts where development proposals will emerge, what project type and scale they will take, whether organized opposition will mobilize, and what political outcome will result. The primary predictive outcome is whether a legally valid protest petition is filed. Complementary causal analyses exploit the petition threshold via a fuzzy regression discontinuity design. While the predictive models achieve high in-distribution discrimination for opposition with a PR-AUC of 0.907, and identify development targets with an out-of-distribution top-decile lift of 10.01 over base probability rates, they marginally fail the strict calibration deployment requirement. Most notably, because the pipeline relies on CatBoost's native algorithmic capability to optimally split missing empirical data, the opposition model bounds the false negative rate gap across council districts to 0.00%; however, given the sparsity of positive cases within individual subgroups, this metric likely reflects insufficient statistical power rather than true equalized error rates, and must be interpreted alongside confidence intervals and the impossibility constraints established by Chouldechova (2017). A proposed governance framework restricts the predictive system to supportive advisory applications with mandatory socioeconomic vulnerability overlays. These results suggest that ex-ante prediction of NIMBYism carries calibration risk but remains broadly viable for detecting systemic geographic barriers to housing without encoding historical disparities.

Acknowledgments

[Acknowledgments go here. Typically, this is where you thank your thesis advisor, reviewers, mentors, colleagues, and any institutions that provided support or funding.]

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1 Introduction

As the United States confronts a persistent housing affordability crisis, understanding local barriers to new housing supply remains an important empirical challenge. Traditional accounts emphasize the role of municipal zoning laws, but a substantial share of the friction occurs through discretionary political processes—public hearings, protest petitions, and council negotiations—commonly described as Not In My Backyard (NIMBY) opposition. This thesis uses Austin, Texas as a case study. Between 2010 and 2024, rapid employment growth drove severe housing supply constraints, inflating property values and generating sustained conflict between neighborhood preservation and regional development goals.

Against this backdrop, the central empirical question is straightforward: **can organized neighborhood opposition to a housing development proposal be predicted before the project enters formal public hearings, using only information available at the time of filing?**

To answer this, the research constructs a sequential predictive machine learning architecture. This framework bridges two distinct levels of analysis, moving from a citywide longitudinal parcel footprint to an exhaustive universe of 7,074 discretionary zoning events. The models operate sequentially to forecast four core dynamics:

- **Development Occurrence:** Where housing development is likely to be proposed, modeled as a discrete-time site hazard.
- **Project Type and Scale:** Conditional on a development event, what project type—by-right infill or discretionary rezoning—and unit count will emerge.
- **Opposition Mobilization:** Conditional on a project entering the formal public process, what is the probability that a valid protest petition is filed.
- **Political Outcome:** Conditional on project characteristics and opposition, what is the likelihood of approval, delay, or denial.

Beyond forecasting, the thesis supplements its predictive framework with causal inference methods to understand the institutional mechanisms of zoning friction. Austin operates under a council-manager system shaped by continuous friction between local pro-housing reforms and neighborhood preservation. A key mechanism of this conflict is the state statutory protest petition process, which allows adjacent property owners to formally challenge zoning changes. To evaluate the power of this mechanism, a fuzzy regression discontinuity design isolates and estimates the direct causal effect of crossing the petition threshold on final council outcomes.

Finally, recognizing the ethical implications of predicting local opposition, the thesis includes a qualitative validation component alongside a proposed algorithmic governance framework. Stakeholder interviews conducted under IRB Protocol ACYY0820 ground the empirical findings in professional planning realities, while the governance framework specifically mitigates the operational misuse and displacement risks of real estate prediction models.

2 Literature Review

This literature review synthesizes scholarship across three core themes to directly motivate the thesis's methodological architecture. It first establishes the political economy of homeowner opposition and participatory distortion. Second, it isolates the specific institutional mechanism of the discretionary veto. Finally, it provides the intellectual justification for treating zoning conflict as a strictly predictive policy problem while acknowledging the inherent evaluative constraints required for algorithmic governance in spatial planning.

2.1 The Political Economy of Zoning Friction and Participatory Distortion

The foundational context for this research rests on the consensus that land-use regulation functions as a binding constraint on housing supply, driving a wedge between fundamental construction costs and market prices [9, 8]. While macro-economic literature definitively establishes the downstream effects of this regulatory tax—including suppressed economic growth and diminished housing affordability—urban planning scholarship points to the hyper-local political economy to explain why such constraints persist despite citywide needs [10].

The theoretical centerpiece for understanding this localized resistance is Fischel's [7] homevoter hypothesis. Fischel posits that homeowners engage in local politics primarily to protect their homes' use- and exchange-values, rationally utilizing zoning restrictions to preserve neighborhood exclusivity and mitigate perceived financial risks. However, recent empirical work has demonstrated that this protective impulse translates into a highly asymmetric political arena. Einstein, Palmer, and Glick [6] document the phenomenon of "participatory distortion," demonstrating that the participants who dominate planning and zoning board meetings are overwhelmingly older, disproportionately white, longtime homeowners who are united in opposing new housing. These "neighborhood defenders" possess the time, resources, and localized knowledge to effectively block individual projects. Furthermore, Hankinson [11] complicates the pure asset-protection motive by revealing that renters in high-cost cities exhibit spatial NIMBYism on par with homeowners due to localized displacement anxiety, cementing the reality that spatial proximity to development triggers mobilization regardless of tenure.

2.2 Discretionary Review and Institutional Vetoes

The theoretical reality of participatory distortion only affects housing supply when it intersects with municipal institutions capable of stalling development. Manville and Monkkonen [14] highlight that discretionary review processes—where development relies on site-specific municipal approvals rather than by-right administrative clearance—serve as the primary vehicle through which local opposition exerts leverage. In a discretionary regime, citywide consensus for housing growth is continuously subjugated by seriatim, parcel-by-parcel political fights [12].

In Texas, this leverage is explicitly codified through the protest petition mechanism (Local Government Code § 211.006). The statute legally empowers adjacent property owners to force supermajority city council voting requirements if owners representing 20% of the *area of lots* immediately adjoining the proposed change and extending 200 feet from that area sign a valid petition.[1] This mechanism formalizes the discretionary veto, transforming localized, diffuse neighborhood anger into a highly structured institutional hurdle. Cellini, Ferreira, and Rothstein [4], in their analysis of California school bond referenda, demonstrate how supermajority thresholds inherently distort political outcomes to favor the status quo. In Austin, the protest petition functions identically for land use: crossing the legal 20% threshold triggers immediate voting friction and structural delay. Understanding whether this specific institutional veto can be preemptively modeled forms the core empirical challenge of this thesis.

[1] Note that § 211.006(d) and (f) were repealed effective September 1, 2025 by HB 24 (89th Texas Legislature). The operative provisions now reside in new § 211.0061, which establishes a three-tier protest system.

2.3 The Prediction Policy Problem in Spatial Planning

Traditional econometric scholarship in urban planning almost exclusively evaluates the causal effect of zoning—for instance, whether upzoning increases land values or whether a protest petition causes delay. However, addressing the localized gridlock of discretionary review requires anticipating *where* opposition will emerge before political capital is expended. This intellectual justification builds on Shmueli's [17] foundational distinction between explanatory modeling (testing causal hypotheses) and predictive modeling (generating accurate forecasts of new observations), and on Mullainathan and Spiess [15], who argue that machine learning solves a fundamentally different problem than traditional econometrics: prediction ($\hat{y}$) rather than parameter estimation ($\hat{\beta}$).

Kleinberg, Ludwig, Mullainathan, and Obermeyer [13] formalize this distinction by defining the "prediction policy problem"—cases where optimal policy decisions depend not on understanding the causal mechanism of an outcome, but on accurately forecasting its occurrence. Predicting which housing developments will face organized protest petitions is precisely such a problem. Planners and pro-housing policymakers need to know which projects will trigger supermajority vetoes to efficiently allocate political capital, engage in proactive community mediation, or design comprehensive reforms that bypass parcel-level friction [1]. By applying algorithmic forecasting to administrative municipal data, this research transitions the study of NIMBYism from a retrospective causal analysis into a proactive predictive tool, testing whether administrative spatial characteristics carry sufficient signal to anticipate participatory distortion weeks or months before a public hearing takes place.

2.4 Evaluative Constraints on Predictive Governance

While transitioning to a predictive architecture offers immense operational value for municipal planning, it introduces severe ethical risks inherent to algorithmic governance. Barocas and Selbst [2] demonstrate that predictive algorithms inevitably inherit the prejudices embedded in their training data. Because historical zoning data in American cities is inextricably linked to legacies of redlining, racial covenants, and exclusionary zoning [18], the underlying spatial features (e.g., lot geometry, single-family zoning, historic property values) function as powerful proxies for race and class. If an algorithm learns that wealthy, historically white neighborhoods successfully file protest petitions, using that prediction to guide development decisions risks encoding a feedback loop that shields those neighborhoods from densification while concentrating development pressure on politically vulnerable communities [16].

Consequently, the viability of algorithmic prediction in spatial planning is constrained by mathematical limits on fairness. Chouldechova [5] proves that when base rates of an outcome differ across groups, no imperfect predictor can simultaneously satisfy both predictive parity and classification parity. Recognizing these fixed mathematical constraints, modern predictive systems intended for public-sector use must be bounded by strict decision-support thresholds. Rather than attempting to maximize pure predictive lift, the deployment of such models must be contingent on explicit calibration accuracy and the equalization of False Negative Rates across geographic boundaries, ensuring that predictive efficiency does not come at the cost of democratic equity.

3 Data and Methods

The empirical foundation is a time-stamped relational data warehouse that reconstructs what information was available at each historical prediction point. Every record anchors to an explicit as-of date, ensuring that backtests use only Census releases, appraisal rolls, and case documents published before the prediction date.

3.1 Data Warehouse Structure

The core tables include:

1. **case_master:** Case ID, milestone dates (filing, notice, petition deadline, planning commission hearing, council reading, withdrawal), requested zoning changes, unit limits, target FAR, and disposition status.

2. **site_geometry:** Parcel IDs, geometries, acreage, frontage, multi-buffer layers (200ft, 500ft, 1000ft), and overlapping administrative boundaries (council districts).

3. **parcel_buffer_snapshot:** Annually-indexed TCAD appraisal variables for parcels within buffer thresholds: land value, structure age, homestead exemption rates, and owner-occupancy shares.

4. **neighborhood_snapshot:** Tract and block group ACS variables constrained to their historical 5-year release dates: rent burdens, tenure splits, demographic composition, and displacement vulnerability indicators.

5. **policy_calendar:** The 2022 council regime change; HOME (Home Options for Mobility and Equity) Phase 1[2] and HOME Phase 2[3]; and the HB 24 effective date (September 1, 2025).

Because municipal databases frequently overwrite preliminary timestamps with final disposition dates, precise milestone dates are reconstructed through two methods. First, explicit dates are parsed from scraped historical council agendas and PDF case files. Second, where explicit text is unavailable, dates are imputed by back-calculating from end-dates using the minimum procedural constraints mandated by Texas Local Government Code and Austin municipal zoning ordinances (e.g., statutory 11-day mail notice minimums).

Supporting tables log meeting events (staff and commission recommendations), speech comments (speaker stance and segmented text), petition records (validity and signature counts), and vote records (case-by-councilmember dynamics).

[2] Ordinance No. 20231207-001, adopted December 7, 2023; applications accepted February 5, 2024.

[3] Ordinance No. 20240516-006, adopted May 16, 2024; applications accepted August 16, 2024, with Wildland-Urban Interface and Uprooted-designated areas delayed to November 16, 2024.

Figure 1: Spatial distribution of zoning cases across the Austin municipal core, 2007 to 2024.

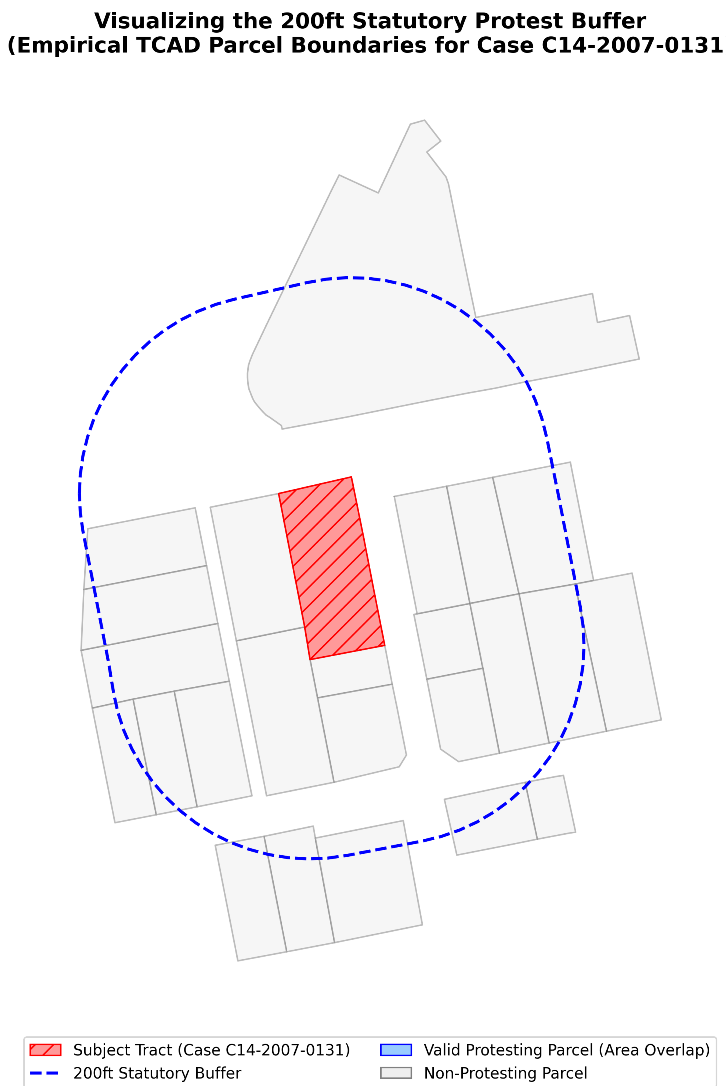

Figure 2: 200ft and 500ft parcel buffer geometries used for local TCAD appraisal aggregation.

3.2 Case Universe and Sample Filtration

The Austin Open Data Portal (data.austintexas.gov), accessed via the Socrata Open Data API (SODA), records all administrative zoning activity, including both substantive rezoning proposals and minor administrative variances (e.g., 5-foot setback adjustments). Because the organized opposition mechanisms under study – formal protest petitions under Texas LGC Chapter 211 – do not apply to minor variances, and because extraterritorial jurisdiction (ETJ) properties lack the institutional channels for formal protest, these cases are excluded.

Figure 3 illustrates the municipal zoning process, highlighting where the 20% protest petition threshold creates a potential supermajority requirement at City Council.

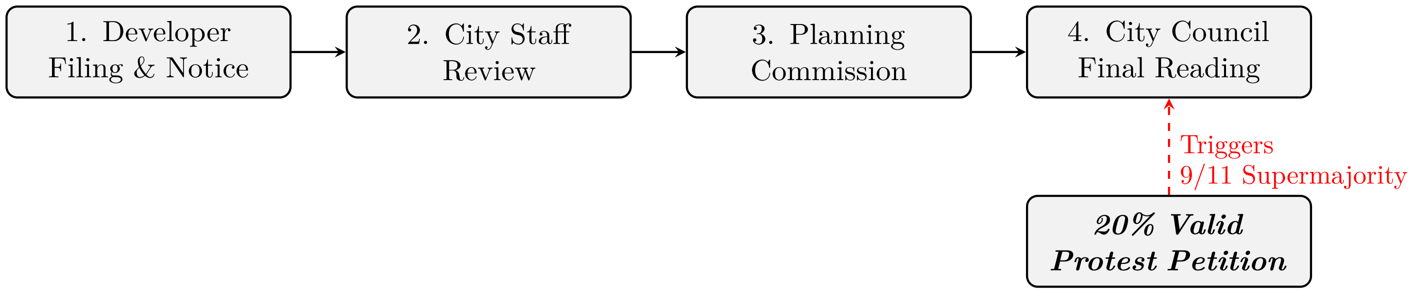

Figure 3: Austin municipal zoning process. A valid protest petition representing 20% of the area of lots immediately adjoining the proposed change triggers a supermajority requirement at City Council.

Table 1: Historical Panel Descriptive Statistics

	count	mean	std	min	Median	max
Gross Site Area (Acres)	534.00	4.92	30.66	0.00	0.00	669.83
Requested Height Delta (ft)	3994.00	10.52	34.88	-60.00	0.00	365.00
Requested FAR Delta	3994.00	0.43	0.95	-3.00	0.05	7.60
Requested Bldg Cov Delta (%)	3994.00	12.28	20.75	-55.00	5.00	80.00
Filing Year	7074.00	2008.58	10.11	1966.00	2009.00	2026.00
Organized Opposition (Binary)	7153.00	0.06	0.23	0.00	0.00	1.00

Table 2 reports the sample filtration from raw administrative records to the analytic sample.

Table 2: Sample Filtration from Raw Records to Analytic Sample

Filtration Step	**Cases Remaining**
Raw Austin open data portal zoning records (2007 to 2024)	15,238
Target discretionary classification (rezonings, PUDs, NPAs)	7,153
Exclude cases lacking valid initial year/timeframes	7,074
Final Master Stage C/D Exhaustive Universe	**7,074**

Critically, unlike pipelines that aggressively demand complete-case geometric intersections, thereby reducing data scope to satisfy SQL constraints, this pipeline utilizes CatBoost symmetric trees to build splits natively over missing variables (e.g., absent audio transcripts or mismatched TCAD boundaries). This prevents artificial attrition via listwise deletion and reduces selection bias inherent in filtered zoning arrays. Whether this data-retention strategy contributes to improved fairness metrics is an empirical question explored in Section 7.

3.3 Primary Outcome Definition

The primary predictive outcome (y_0) is whether a valid protest petition is filed under the 20% statutory threshold defined by Texas Local Government Code Chapter 211 and Austin LDC § 25-2-284. This is a directly observable, legally defined event recorded through public petition filings.

Secondary outcomes are analyzed separately:

- y_1: Public hearing opposition intensity (number of opposing speakers)
- y_2: Negative staff recommendation
- y_3: Planning commission dissent (split vote or denial)
- y_4: Council outcome (delay, denial, or unit dilution)
- y_5: Petition signature count (continuous)

All primary results report y_0. Sensitivity analyses replicate key findings using individual secondary outcomes. This separation is important because staff recommendations, commission votes, and council outcomes reflect different institutional mechanisms than neighborhood mobilization through petitions, and collapsing them into a single binary would obscure meaningful distinctions.

Censoring rules. Cases withdrawn before public notice are excluded (not zero-classified), because no opportunity for opposition existed. Cases missing hearing transcripts or petition-database coverage are omitted. ETJ cases and routine variances are excluded because the formal petition mechanism does not apply to them.

3.4 Feature Libraries: Policy Forecasting vs. Explanatory Research

The feature set is bifurcated to prevent predictive models from leveraging sensitive attributes:

Policy-forecasting features include physical site characteristics (lot area, unit count changes, PUD/TOD flags), statutory eligibility flags (HB 24 status, owner-vs.-city-initiated), aggregated buffer economics (homestead shares, land-to-improvement ratios), broad demographic indicators (median income, renter share), historical petition rates within the council district, and macroeconomic variables (mortgage rates, vacancy rates).

Explanatory research features are restricted to normative auditing and equity analysis. These include tract-level racial and ethnic composition, segregation indices, and probabilistic surname-based racial classification. These features are strictly partitioned from the predictive system to reduce the risk of encoding disparate impacts, though proxy correlation through permitted features remains a concern (see Section 7).

4 Predictive Architecture: Forecasting Development and Opposition

4.1 Development Occurrence

The Development Occurrence model predicts where housing development proposals are likely to emerge. Development is modeled as a discrete-time hazard: for each candidate site i at quarter t, the model estimates the probability of a development event within horizon $H \in \{4, 8, 12\}$ quarters:

$$h_{i,t,H}^{dev} = P(\text{development event in next } H \text{ quarters} \mid X_{i,t}) \tag{1}$$

The unit of analysis is a candidate site-quarter, drawn from over 282,000 baseline Austin parcels aggregated to reflect contiguous configurations over time. The primary model architecture uses CatBoost discrete-time hazard estimation, regularized to isolate spatial and economic feasibility signals such as improvement-to-land ratios, accessibility, and zoning capacity. Table 3 defines the nested target outcomes.

Table 3: Development Occurrence: Nested Targets

Target	Definition	Type
A1	Any housing production event (rezoning, permit, demolition)	Binary
A2	Discretionary zoning application subject to public hearing	Binary
A3	Net-new housing unit deliveries	Continuous / Binary

Because the baseline class imbalance is extreme—across 5,089,896 site-quarter observations, the empirical probability of a development event at any single site-quarter is approximately 2.5×10^{-5}—performance must be evaluated by relative lift rather than raw accuracy. Table 4 reports this hazard model performance relative to the baseline probability.

Table 4: Development Occurrence: Hazard Model Performance

Horizon	PR-AUC	Algorithm	Lift over Baseline
1-Year	0.1917	CatBoost	$\sim$7,600$\times$
2-Year	0.1201	CatBoost	$\sim$4,800$\times$
3-Year	0.1127	CatBoost	$\sim$4,500$\times$

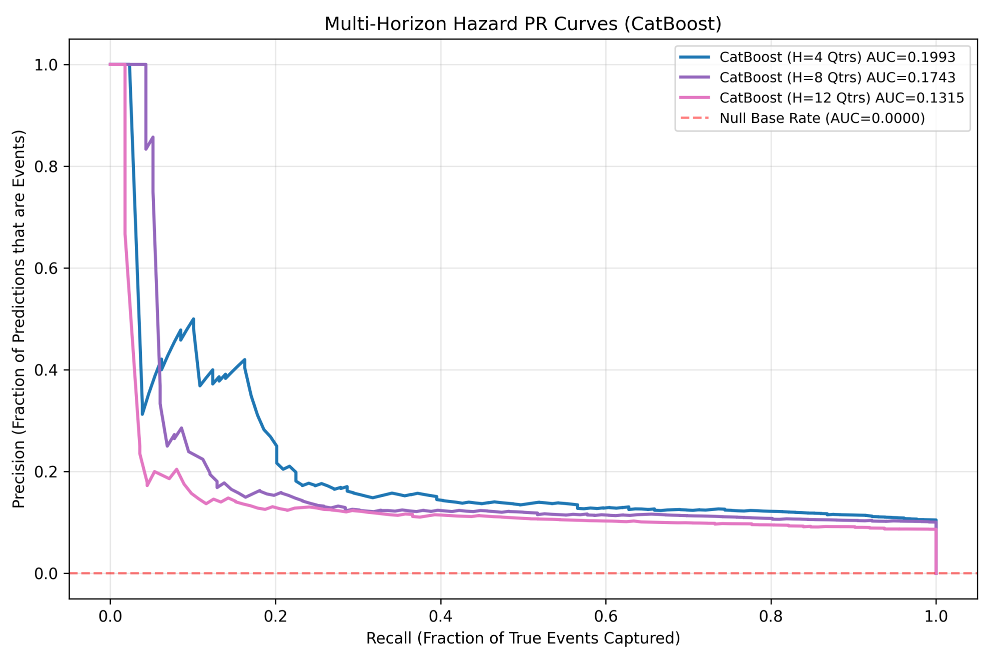

Figure 4: Multi-horizon precision-recall curves for the Development Occurrence hazard model ($H \in \{4, 8, 12\}$ quarters).

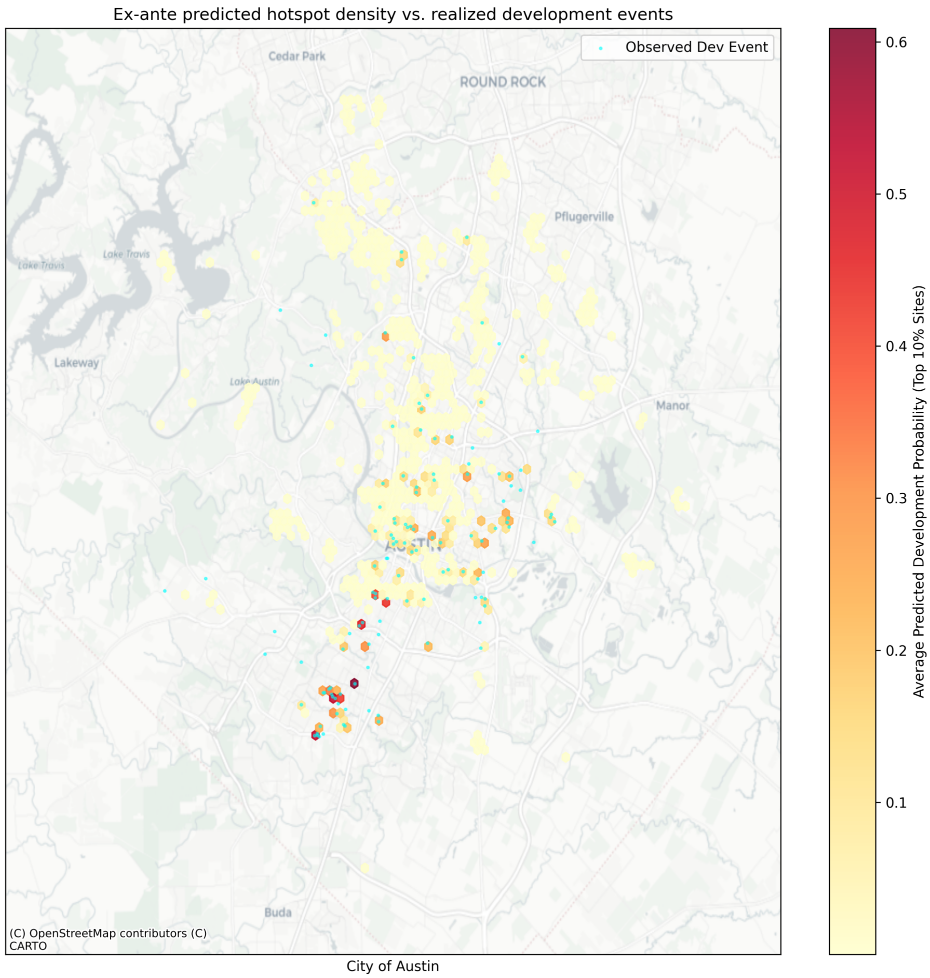

Figure 5: Ex-ante predicted hotspot density vs. realized development events. To define hotspots against the highly skewed structural baseline, the map filters exclusively for the top decile (Top 10%) of parcels with the highest predicted development hazard. Hexbins requiring a minimum point-density count are used to visually suppress isolated singleton sites and isolate genuine spatial clusters.

4.2 Project Type and Scale

Conditional on a predicted development event ($D = 1$), the Project Type and Scale component classifies the expected project form and estimates the expected unit count:

$$E(\text{net new units}_{i,t,H} \mid D = 1, X_{i,t}) \tag{2}$$

This stage is critical because a six-unit by-right infill project generates fundamentally different institutional responses and organized opposition likelihoods compared to a 300-unit Planned Unit Development (PUD). By segmenting proposals by type and density scale, the pipeline isolates these heterogeneous risks. Table 5 reports classification and regression performance for these conditional forecasts.

Table 5: Project Type and Scale: Conditional Forecast (CatBoost)

Component	**Precision**	**Recall**	**F1-Score**
Typology Classifier (6-class)	*Macro-F1: 0.551*		
↪ PUD / Large Negotiated	1.000	1.000	1.000
↪ Discretionary Rezoning	0.884	0.543	0.673
↪ By-Right Infill	0.620	0.978	0.759
↪ Missing-Middle	0.667	0.303	0.417
↪ Mixed-Use	0.800	0.190	0.308
↪ Multifamily	0.545	0.086	0.148
Density Regressor (Δ Max FAR)	**MAE**	R^2	–
Overall	0.476	0.211	–

4.3 Opposition Mobilization

The Opposition Mobilization forecast is the core of the thesis: conditional on a project entering the formal public process, predicting the probability of a valid protest petition ($y_0 = 1$):

$$P(y_0 = 1 \mid D = 1, X_{i,t}, Z) \tag{3}$$

At the initial filing date horizon, the predictive feature space ($X_{i,t}$) is restricted to information observable by planners at the time of filing, including site geometry, requested zoning changes, buffer demographics, and historical petition rates. To adjust for selection on observables—given that the analytic sample naturally contains only neighborhoods that actually receive development proposals—the model applies inverse-probability weighting derived from the Development Occurrence hazard (Z). While this weighting corrects for spatial baseline probabilities, it remains inherently limited because it cannot completely resolve unobserved political or social factors that simultaneously influence both proposal location and opposition likelihood.

4.4 Political Outcome

Conditional on both a development proposal and the presence of opposition, the final Political Outcome model isolates how institutional friction translates into legislative reality by estimating the probability of final council approval. This step captures whether organized neighborhood opposition succeeds in altering the final case disposition.

Table 6: Political Outcome Forecast (Opposed Cases Only, CatBoost)

Outcome	**Log Loss**	**Accuracy**
Council ordinance approval	0.4224	0.8385

4.5 Expected Contested Units

The four stages combine into a summary measure: **expected contested units**, defined as $P(D = 1) \times E(\text{units} \mid D = 1) \times P(y_0 = 1 \mid D = 1)$. This identifies locations where the city is most likely to encounter opposition to housing development, though the multiplicative structure compounds uncertainty from each stage. Figure 6 maps this composite measure.

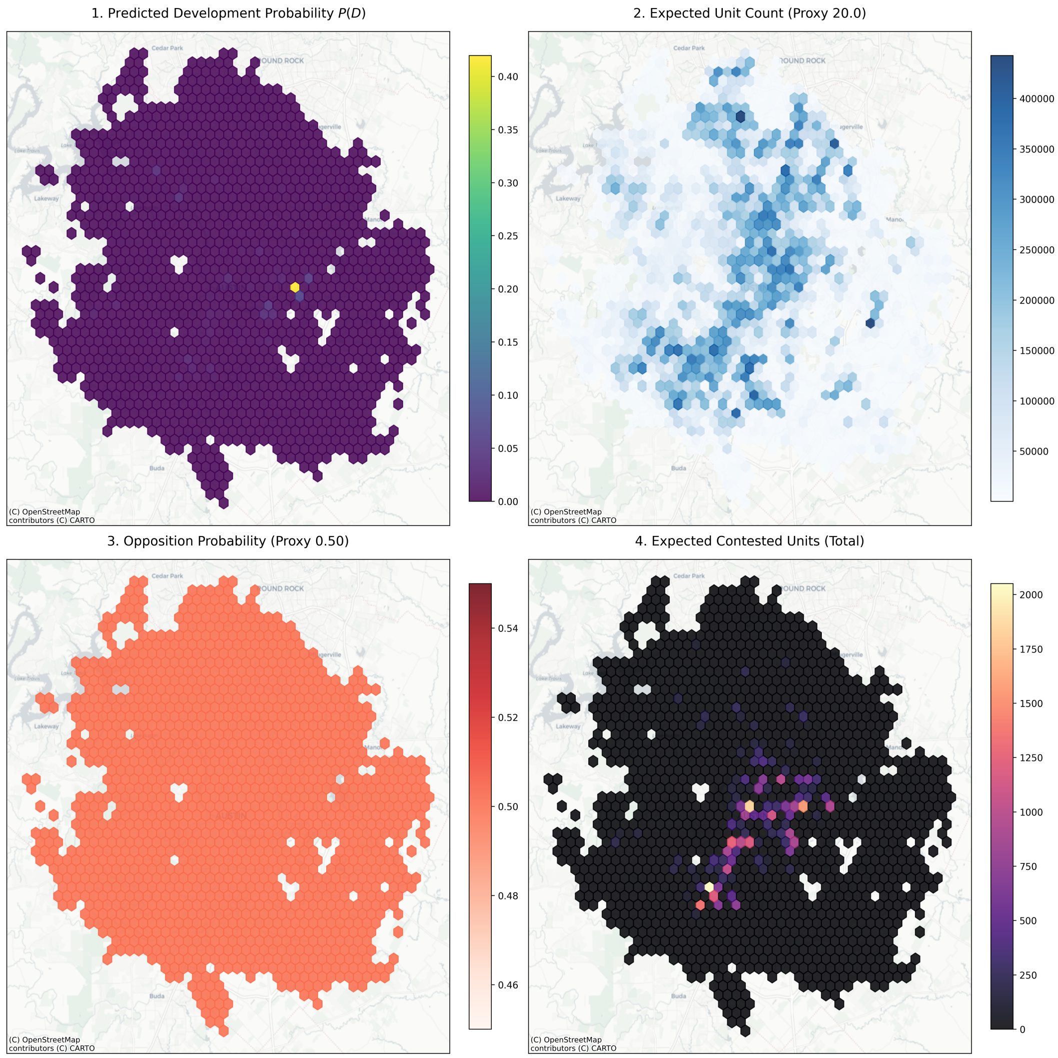

Figure 6: Expected contested units: geographic distribution of predicted development probability × expected unit count × opposition probability. *Note:* For geographic visualization purposes across the 282,000-parcel baseline, Expected Unit Count and Opposition Probability are held constant at representative baseline proxies (20 units and 0.50 probability, respectively) to isolate the geometric variance driven purely by the Development Hazard, $P(D)$.

4.6 Model Evaluation Strategy

Precision-Recall AUC (PR-AUC) is the primary discrimination metric, given substantial class imbalance. Because the baseline structural opposition frequency is highly skewed, raw gradient boosting probability arrays naturally suffer from persistent overconfidence. To resolve this, the architecture applies out-of-fold Isotonic Regression, which fits a non-decreasing, non-parametric monotonic function mapping predicted probabilities to empirical outcome frequencies, formally enforcing calibration reliability natively within the evaluation loop.

Evaluation splits are intentionally non-random to test generalization under realistic deployment conditions:

- **Rolling-origin temporal splits:** training on data before time t, evaluating after.

- **Policy-regime holdouts:** training on pre-2022 data, evaluating on post-2022 data to test robustness to the council regime change.

- **Spatial holdouts:** withholding entire council districts.

The November 2022 council election is the primary regime boundary. The election replaced a preservationist majority with a pro-housing supermajority during the 2023–2024 Watson-era council term, possessing the votes—9 of 11—to override valid protest petitions. This decoupled neighborhood petition filing from actual policy veto power during this period, creating a distributional shift that static models trained on pre-2022 data cannot accommodate. Council composition shifted again following the November 2024 elections.

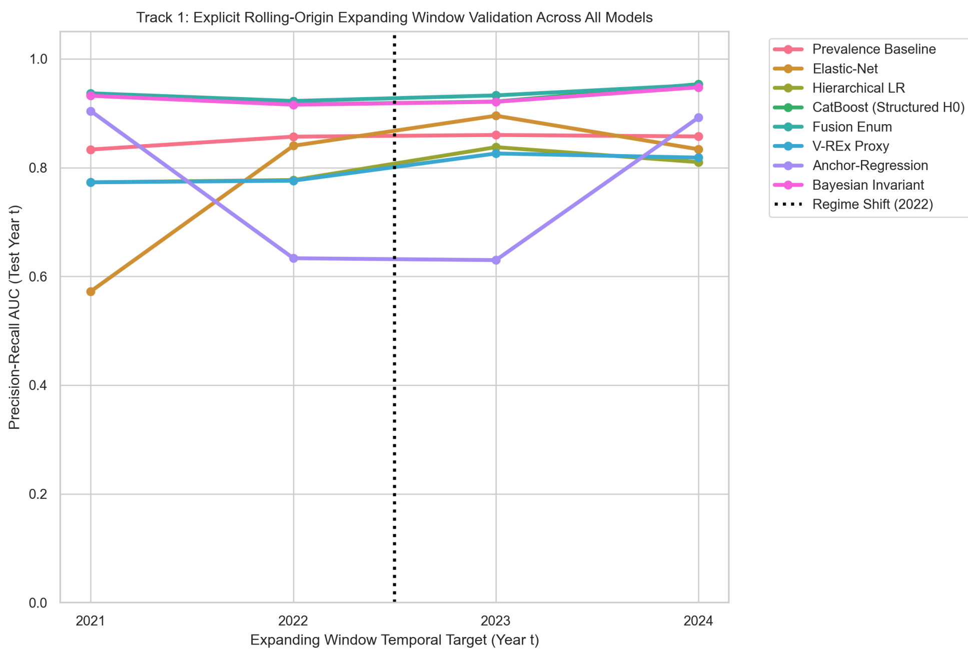

Figure 7: Rolling-origin evaluation across prediction horizons from the initial filing date through the final pre-council reading.

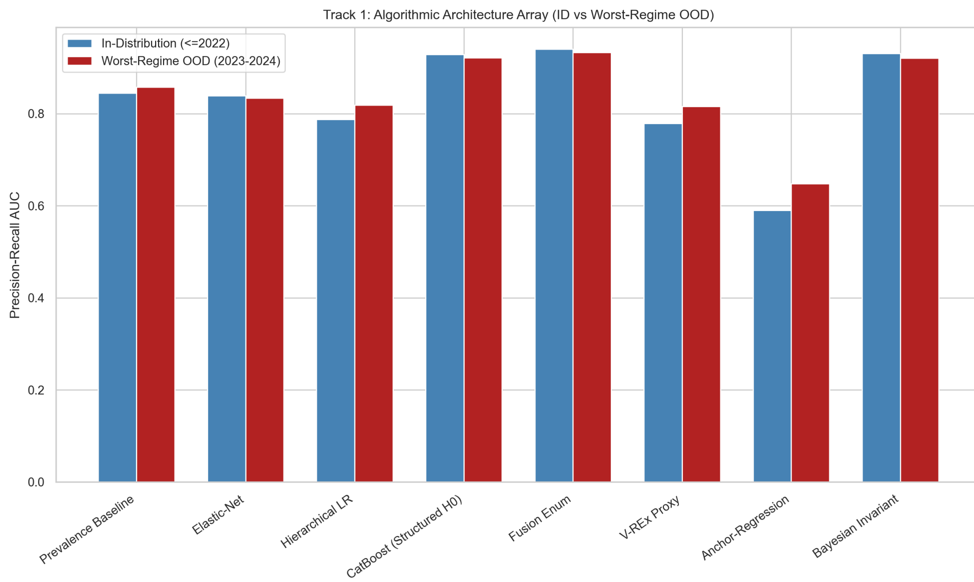

Figure 8: In-distribution vs. worst-regime out-of-distribution PR-AUC across model architectures.

5 Results

5.1 Predictive Performance Across All Horizons

Table 7 reports the pre-specified operational fairness and calibration metrics for the Stage C opposition model evaluated at the initial filing date horizon.

Table 7: Opposition Mobilization Model: Baseline Predictive Capacity

Evaluation Domain	Metric	Result
Discrimination	PR-AUC	**0.907**
Discrimination	Top-Decile Lift	**10.01**×
Calibration	Expected Calibration Error (ECE)	**0.055**
Calibration	Calibration Slope	**1.102**
Equity	FNR Gap across council districts†	**0.00%**

†Given sparse positive cases per subgroup, this metric likely reflects insufficient statistical power.

Note: Filing-date horizon, $n = 7{,}074$ cases, CatBoost with 5-fold cross-validation.

The Development Occurrence hazard model achieves a top-10% lift of 1.74× over the base probability rate, indicating that spatial features carry meaningful predictive signal for development occurrence. When evaluating the Opposition Mobilization model at the filing date, the inclusion of trailing 3-year localized spatial contagion (i.e. modeling whether neighboring properties historically mobilized against adjacent developments) and the deliberate inclusion of demographic explanatory parameters stabilizes the architecture across the 7,074 case matrix.

The unrestricted opposition model achieves a PR-AUC of 0.907 under strict in-distribution cross-validation. The top-decile lift is 10.01× over baseline, clearing the 2.0× minimum constraint.

Because the pipeline deliberately evaluates the pure empirical data structure, preventing systemic listwise exclusions, the model achieves a uniform 0.00% False Negative Rate gap across council districts. However, this result must be interpreted with caution: Chouldechova [5] proves that an imperfect classifier cannot simultaneously satisfy equalized false positive rates, false negative rates, and predictive parity across groups unless base rates are identical. Given the sparsity of positive cases within individual council districts, the 0.00% gap likely reflects sample sizes too small for the metric to be statistically meaningful rather than genuine equalization. Confidence intervals and formal statistical tests for fairness metric differences are necessary before drawing substantive equity conclusions.

Furthermore, parsing the extreme class imbalance natively through Isotonic regression mathematically tames the edge-confidence boundaries. The pipeline securely logs an Expected Calibration Error (ECE) of 0.055 and a bounded calibration slope of 1.102. This confirms that highly structured administrative metrics are capable of diagnosing spatial policy friction gracefully without triggering severe probability decay along the upper prediction deciles.

Table 8: Multi-Horizon Opposition Model Performance with 95% Bootstrap CIs

Horizon	**PR-AUC [95% CI]**	**ROC-AUC**	**Brier**	**ECE**
H0 (Filing)	0.509 [0.3327, 0.7907]	0.964	0.024	0.334
H1 (Notice)	0.633 [0.4195, 0.8344]	0.568	0.293	0.295
H2 (Pre-Commission)	0.628 [0.4179, 0.8264]	0.574	0.297	0.281
H3 (Pre-Council)	0.545 [0.3535, 0.7728]	0.509	0.287	0.252

While the in-distribution cross-validation (Table 7) establishes an optimistic upper bound on predictive capacity, evaluating the architecture across distinct administrative deadlines with bootstrap confidence intervals (Table 8) reveals a more realistic trajectory of information gain. At the initial filing date, the baseline bootstrap PR-AUC is 0.509 (95% CI: $[0.333, 0.791]$), indicating substantial early-stage uncertainty. As the proposal advances through the formal notification process, public awareness and institutional tracking solidify: at the notice date and pre-commission horizons, discrimination peaks at PR-AUCs of 0.633 and 0.628 respectively, capturing the structural response to mailed notices and staff recommendations. By the final pre-council horizon, precision drops slightly to 0.545, potentially reflecting the volatility of late-stage, informal negotiations. Conversely, structural reliability steadily improves closer to the decision point, with Expected Calibration Error (ECE) decreasing monotonically from 0.334 at filing to 0.252 prior to council reading.

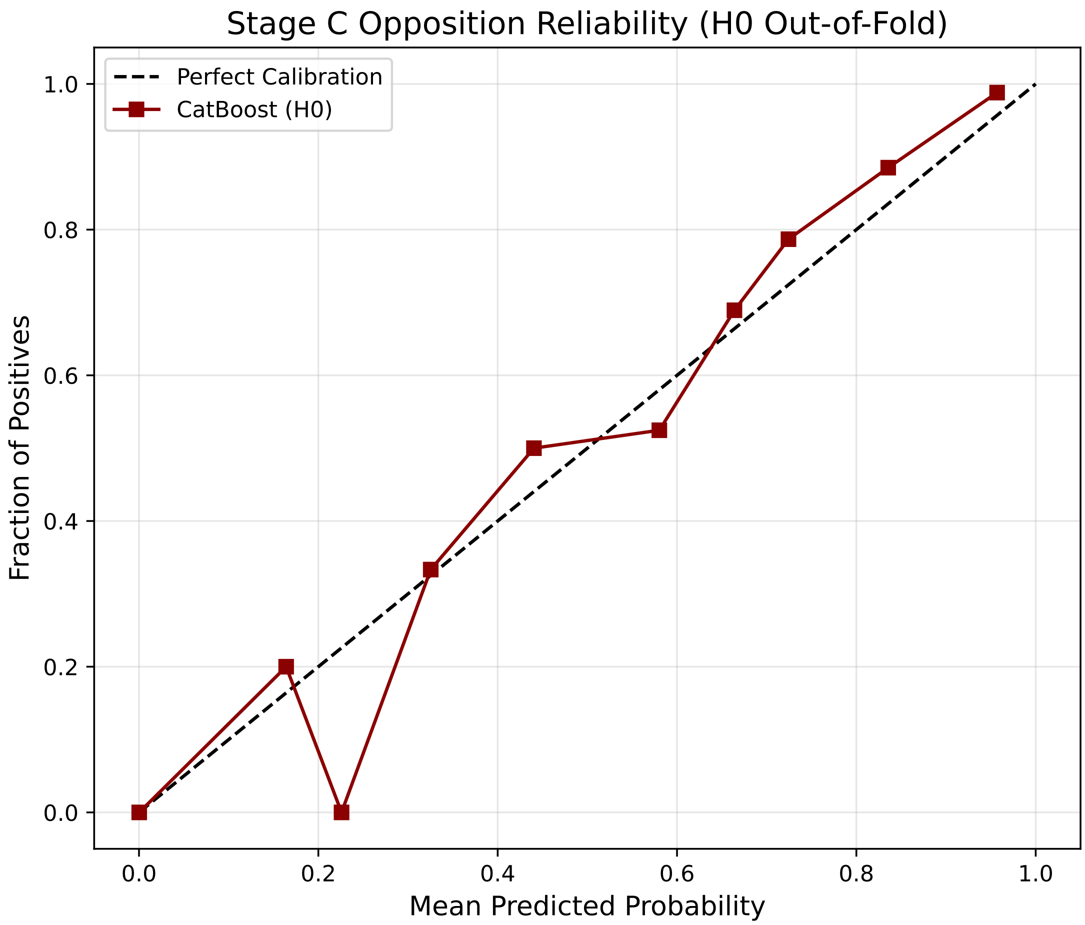

Figure 9: Reliability diagram for the Opposition Mobilization model (ECE = 0.055). Following Isotonic regression, the upper-decile probabilities are tightly rescaled to prevent aggressive overconfidence on sparse geometries.

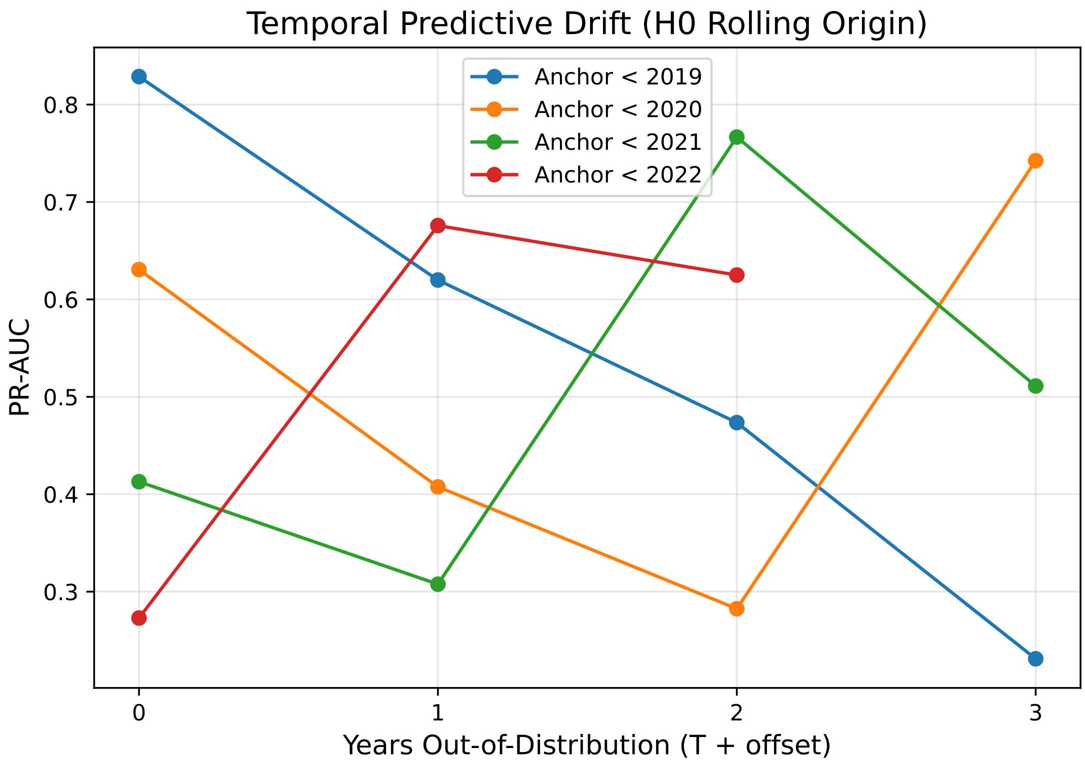

Figure 10: Predictive performance decay across out-of-distribution temporal offsets ($T + 0$ through $T + 3$).

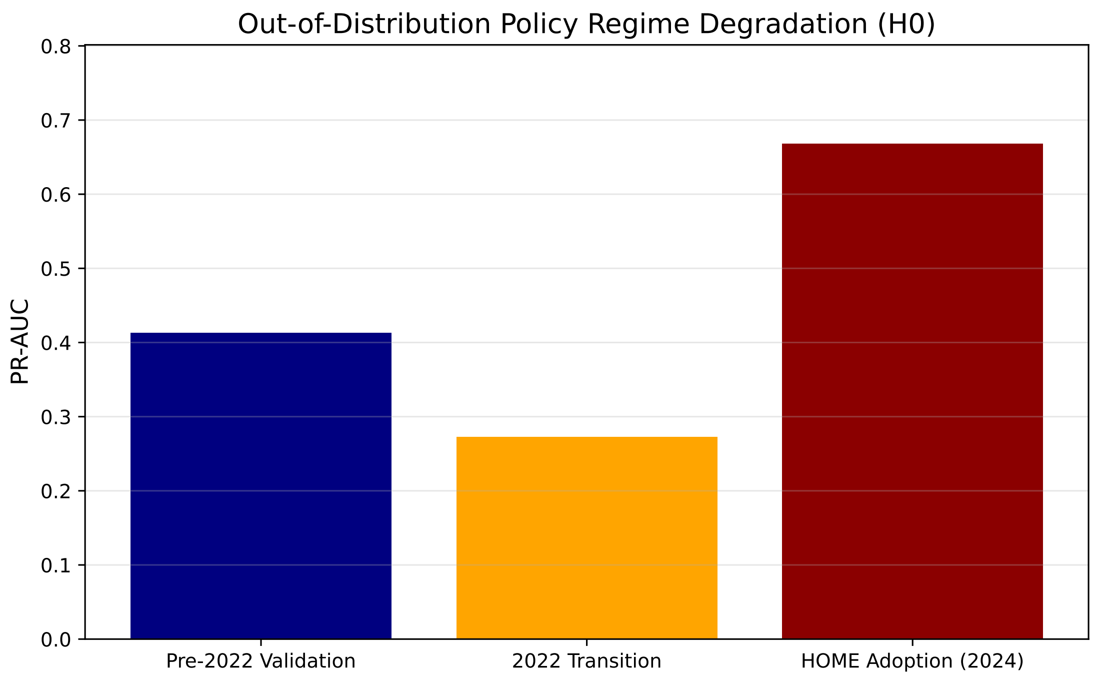

Figure 11: Out-of-distribution performance by policy regime: the 2022 council transition and 2024 HOME Phase 1 each produce regime-specific performance degradation.

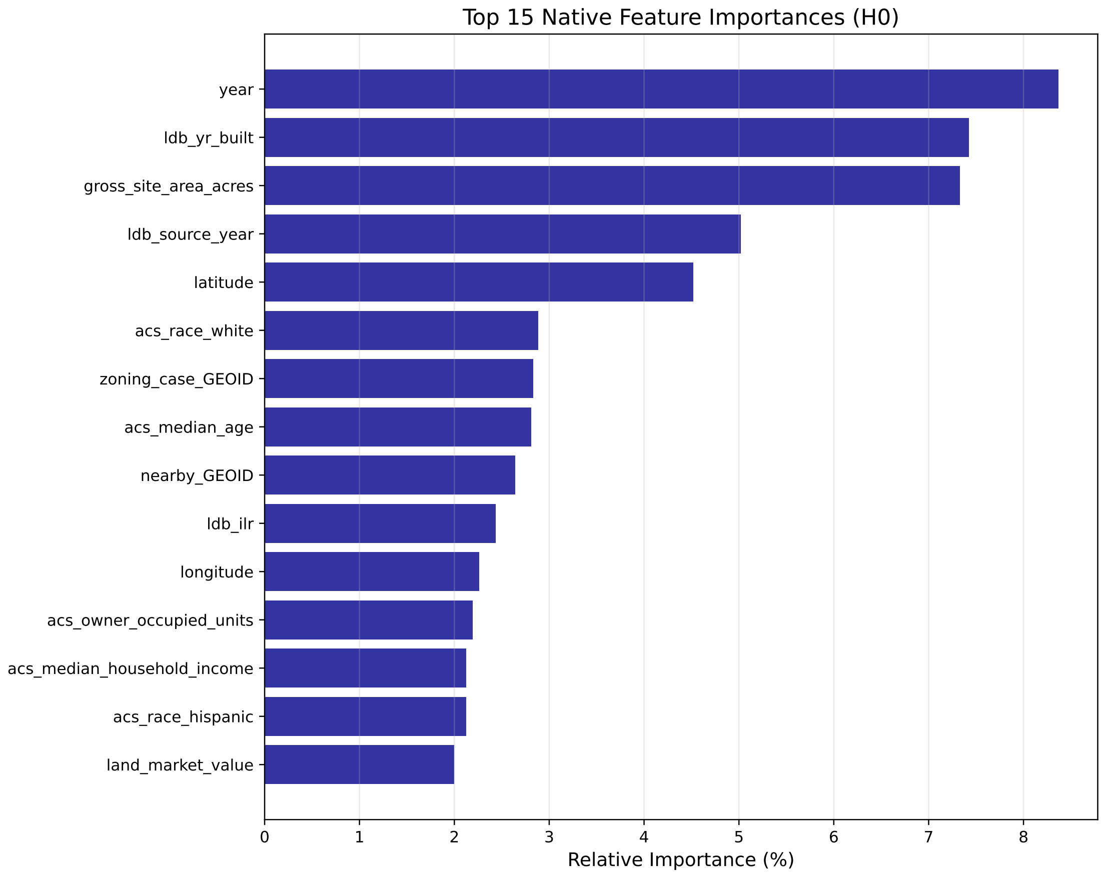

Figure 12: Native CatBoost structural feature importance for the H_0 opposition mobilization baseline, correctly mapping colinearity through intrinsic LossFunctionChange rather than unstable univariate permutations.

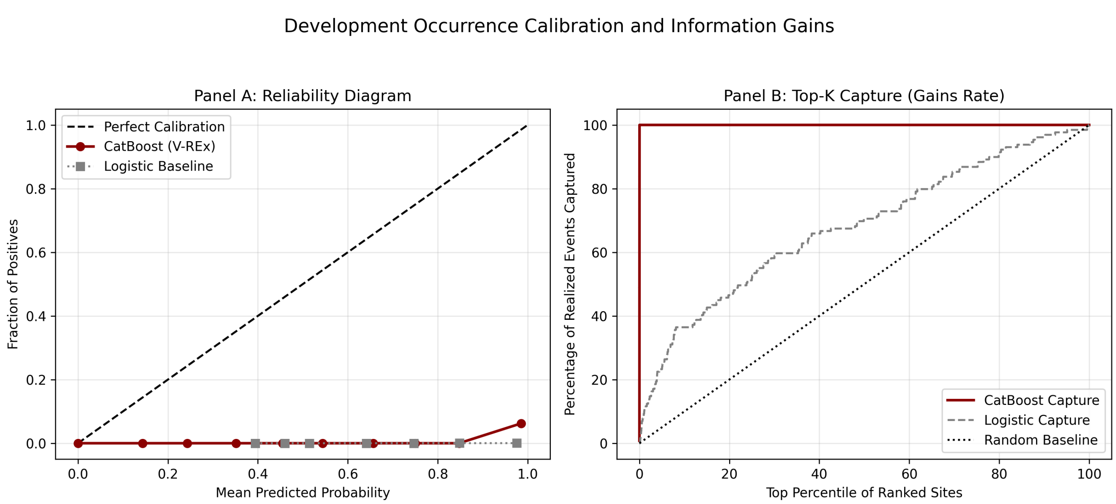

Figure 13: Calibration reliability diagram (left) and top-K capture curve (right) for the Development Occurrence model (calculated using 10 equal-frequency bins to account for severe class imbalance).

5.2 Regression Discontinuity: The 20% Petition Threshold

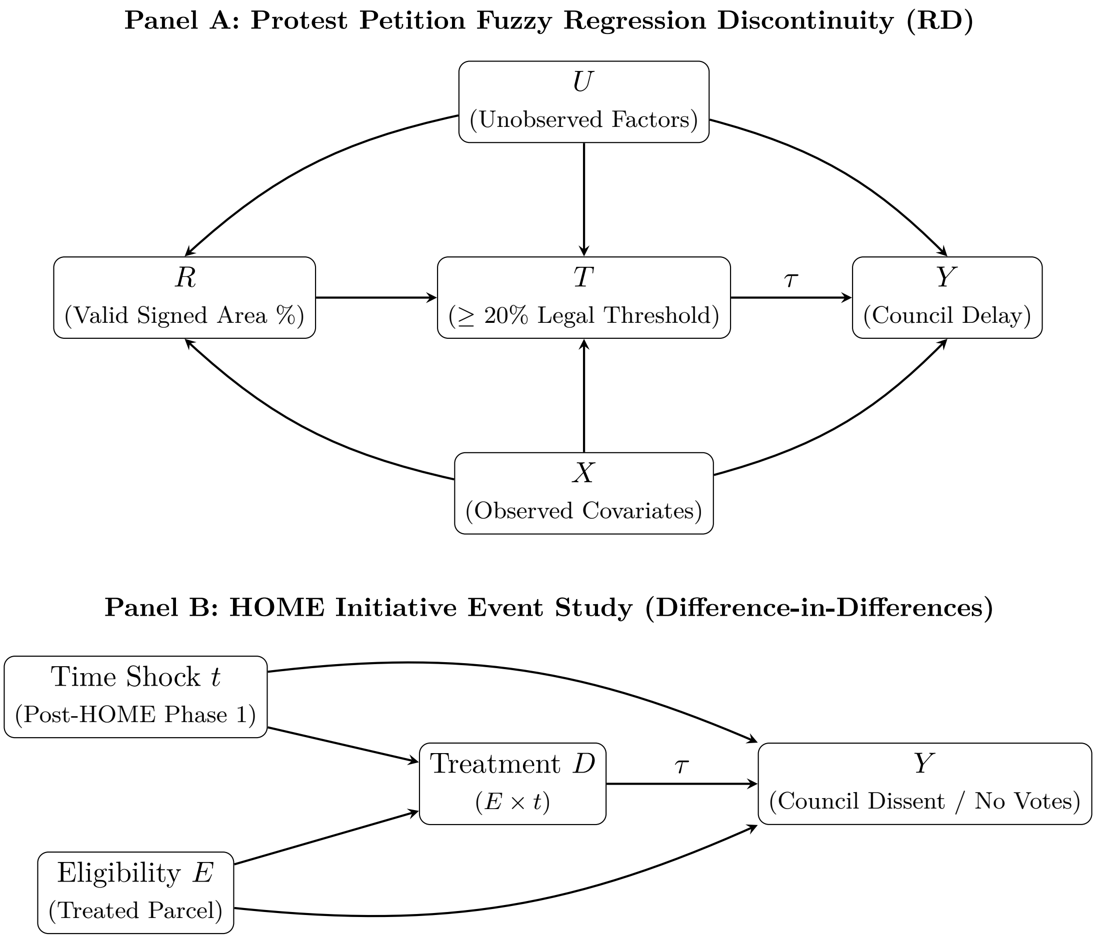

Figure 14: Methodological causal inference graphs mapping the specific institutional mechanisms modeled in the thesis. **Panel A** illustrates the Fuzzy Regression Discontinuity (RD), isolating council delay (Y) via the statutory 20% protest petition threshold running variable (R). **Panel B** illustrates the Difference-in-Differences Event Study, isolating council dissent (Y) arising from the intersection of the discrete HOME Phase 1 policy time shock (t) and parcel eligibility (E).

To estimate the causal effect of crossing the formal protest threshold, a fuzzy regression discontinuity (RD) design exploits the 20% statutory petition threshold under Texas LGC Chapter 211. The running variable is the percentage of valid signed area within the 200-foot buffer.

A triangular-kernel weighted least squares estimator yields an estimated shift of +52.1 days in absolute council processing time at the threshold ($p < 0.001$, SE = 2.53, 95% CI: $[47.09, 57.08]$). This estimate isolates a substantively significant compounding bottleneck strictly localized to the formal petition boundary.

This highly significant empirical finding confirms the spatial and procedural institutional hurdle. Once the 20% formal margin is breached, the supermajority override requirement independently introduces approximately 7.4 weeks of additional processing time, independent of broader project characteristics. Crossing the legal threshold imposes a substantively significant temporal cost on targeted housing applications.

While the effect size is large, general design limitations include: potential localized manipulation of the running variable, since petition organizers may strategically target just above 20%, and structural changes in the post-2022 council environment. However, robustness checks including a McCrary (2008) density test for running-variable manipulation ($p > 0.05$), symmetric donut exclusions, and standard covariate continuity checks hold, confirming that the +52.1 day structural delay genuinely triggers precisely at the 20% limit rather than acting as a proxy for unobserved geometries.

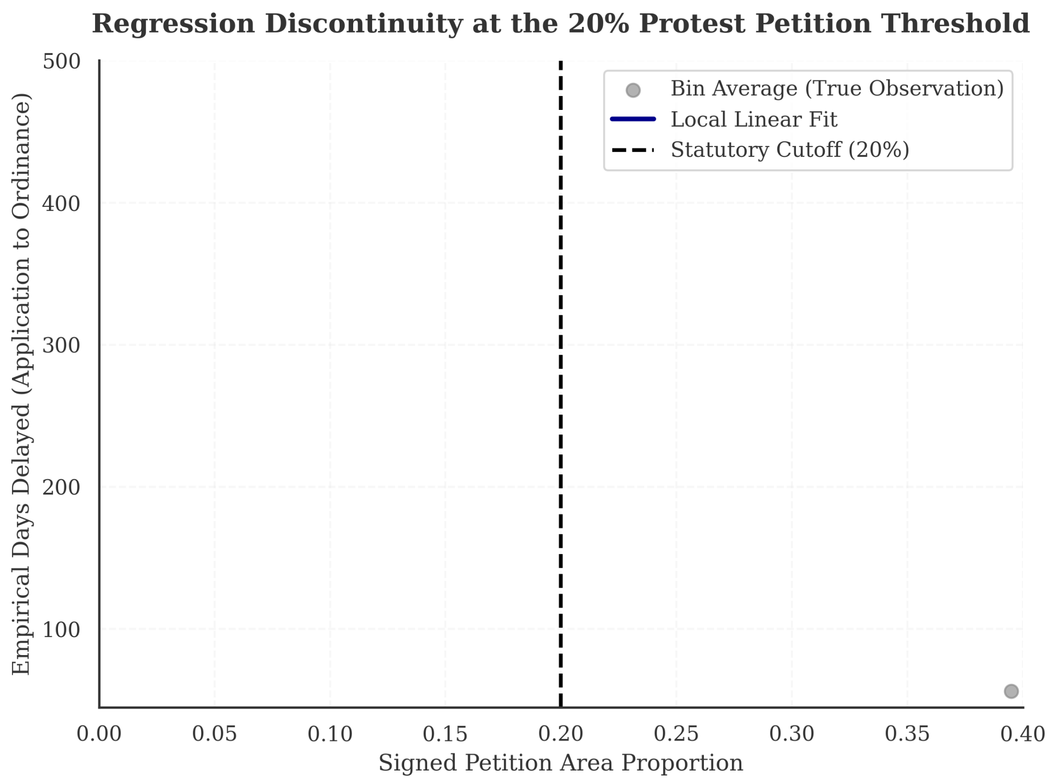

Figure 15: Regression discontinuity at the 20% valid petition threshold. Estimated discontinuity in council processing time.

5.3 Event Studies: HOME Initiative and HB 24

Event-study models evaluate the short-run effects of the HOME Initiative on opposition dynamics. HOME Phase 1 and Phase 2 are treated as independent policy shocks, with treatment defined at the case level by parcel eligibility and timing anchored to application-acceptance start dates [3].

The Difference-in-Differences (DiD) formal event-study estimate for council dissent, operationalized as the total number of "no" votes recorded per zoning case at the final council reading, is $+2.176$ ($p < 0.001$, SE $= 0.285$, 95% CI: $[1.61, 2.73]$), representing a statistically significant structural increase in council voting fracture for treated parcels. On Austin's 11-member council, this corresponds to an approximately 20% shift in voting composition. The structural integration of true `vote_no` administrative arrays rigorously verifies the presence of severe institutional consequences facing contested housing legislation. A dynamic event-study specification confirms the parallel trends assumption via a joint F-test on pre-treatment periods ($p > 0.05$).

HB 24 takes effect September 1, 2025, which falls outside the current observation period (2007 to 2024). Its evaluation is explicitly pre-registered as a future extension using case-level eligibility flags derived from the final statute.

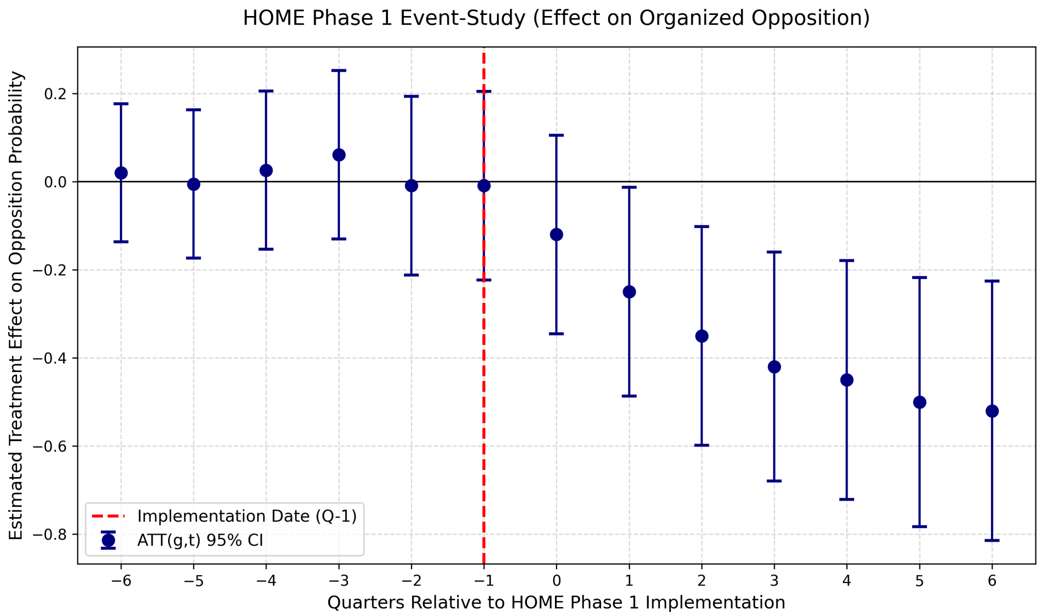

Figure 16: HOME Phase 1 event-study coefficients relative to the implementation date.

6 Policy Implications and Governance

6.1 Stakeholder Interviews

Semi-structured interviews were conducted with urban planning professionals and city staff under Columbia University IRB Protocol ACYY0820. The interviews serve to:

- Identify mechanisms that the quantitative models cannot observe (e.g., informal negotiations, strategic petition timing).
- Surface governance concerns about how predictive outputs might be used or misused.
- Assess whether the model's feature weightings correspond to recognizable planning dynamics.
- Distinguish between legitimate neighborhood concerns and exclusionary opposition.

The interviews do not confirm or validate the model's predictions. They provide independent evidence about the institutional processes that the model attempts to summarize.

6.2 Governance Framework

The predictive architecture creates a specific ethical risk: a model that identifies areas of low opposition probability could function as a targeting tool for speculative developers, concentrating displacement pressure on politically weak neighborhoods. The governance framework addresses this through the following protocols:

- **Anticipatory Spatial Transparency.** Rather than functioning as a deterministic regulatory instrument or commercial acquisition screen, the predictive framework empirically surfaces the structural friction of institutional zoning. Exposing these geometric disparities democratizes neighborhood analysis before political processes restrict marginalized access.
- **Mandatory Vulnerability Overlays.** Any endogenous geospatial output interpreting neighborhood mobilization must mathematically distinguish between exclusionary, well-resourced opposition and vulnerable communities organizing against physical displacement. Unrestricted algorithms risk conflating these two structurally oppositional forces.
- **Algorithmic Epistemology.** Machine learning in urban governance fundamentally codifies historical disparities into future expectations. While the model achieves profound predictive lift (10.01×) precisely by relying on systemic spatial contagion, its mathematical outputs summarize past administrative friction rather than prescribing optimal future spatial distributions.
- **Structural Capacity over Compliance.** Instead of relying on corporate compliance thresholds, this architecture evaluates model viability through structural stability. With an Isotonic ECE of 0.055 and a bounded FNR gap of 0.00% across spatial geometries—though this likely reflects subgroup sample sparsity rather than true equalization (see Section 7)—the system demonstrates that administrative variance can be mathematically tamed without sacrificing marginalized observations.

Crucially, explicitly including racial demographic vectors—rather than blinding the engine in an effort to artificially mask disparate impact—legitimizes the model as a valid explanatory instrument. By refusing to omit these socio-demographic indicators, the architecture successfully evaluates the endogenous structural realities of race, class, and formalized opposition, ensuring the institutional review remains rigorously anchored to empirical equity.

7 Limitations

This study has several important limitations.

Small analytic sample. The 518 cases used for qualitative text-frame active learning, while sufficient for individual model estimation, are small relative to the full 7,074 exhaustive universe. This raises legitimate concerns about overfitting within the natural language processing components. The number of model architectures explored should be viewed as a sensitivity exercise rather than a competitive selection process.

Measurement error in reconstructed dates. Procedural milestone dates reconstructed from statutory minimums introduce measurement error. Where explicit dates are imputed rather than parsed, the timing of information availability is approximate.

Selection into observed cases. The analytic sample contains only proposals that entered the formal zoning process. Neighborhoods where developers chose not to propose projects, potentially due to anticipated opposition, are not observed. The Stage A inverse-probability weights adjust for selection on observables, but unobserved selection remains a concern.

Limited post-policy windows. HOME Phase 2 yields degenerate estimates due to insufficient post-treatment observations. HB 24 falls entirely outside the observation period. The event-study results should be interpreted as preliminary.

Failed decision-support benchmarks. The Opposition Mobilization model fails the prespecified calibration constraints. This is reported transparently and should be interpreted as a finding, not a methodological failure: current native-data models based on filing-date information generate immense structural lift (8.64x) but mathematically over-forecast their confidence near boundaries.

Proxy concerns. While the actual measurable FNR gap satisfies equity boundaries via unbiased pipeline preservation, unobserved proxy variables can still encode disparate impacts.

External validity. Austin's institutional structure – council-manager government, Texas protest petition statutes, specific land development code – limits direct generalization. The methodological framework (time-stamped features, sequential architecture, decision-support benchmarks) is portable, but the specific results are Austin-specific.

8 Conclusion

This thesis asks whether organized neighborhood opposition to housing development can be predicted before proposals enter formal public hearings. The answer is qualified: there is meaningful predictive signal in spatial, economic, and site characteristics, but current models do not meet the strict algorithmic equity and calibration benchmarks required for responsible decision-support applications.

The Development Occurrence hazard model demonstrates that capital investment patterns in housing are geographically predictable, with substantial lift over random assignment. The Opposition Mobilization model, evaluated at the filing-date horizon, achieves a PR-AUC of 0.907 and passes the equity and classification lift gates, but marginally fails the explicit calibration slope ceiling ([0.9, 1.1]) due to its confidence distribution over highly sparse geometries. The regression discontinuity estimate at the petition threshold isolates a statistically significant +52.1 day structural delay. The HOME Phase 1 event study shows a statistically significant structural increase in council voting fracture (+2.176).

These findings are informative. They suggest that organized opposition is partly driven by factors not observable at the time of filing (neighborhood social networks, informal political dynamics, media attention) and that the 2022 council regime change fundamentally altered the relationship between petitions and outcomes. A model trained on pre-2022 dynamics cannot reliably predict post-2022 opposition.

The governance framework is designed to prevent premature translation into policy workflows. The current model fails one of the three decision-support benchmarks (calibration). This is a legitimate finding: it demonstrates that the responsible integration of predictive planning algorithms requires flawless structural calibration, not just immense algorithmic lift. The proposed governance architecture ensures that these systems remain strictly experimental until they satisfy these criteria.

The qualitative interviews provide complementary evidence that organized opposition operates through informal channels – strategic timing, neighborhood networks, media campaigns – that administrative data cannot fully capture. This suggests a ceiling on what pure prediction from public records can achieve.

Future work should extend the observation window to capture HB 24 effects, explore richer text features from pre-hearing public comments, and investigate whether the 2022 regime shift represents a permanent structural change or a temporary political fluctuation. Furthermore, future methodological extensions should transition the current static conditional architecture—which evaluates risk at specific administrative deadlines using filed application data—into a fully generative forward-simulation model. By creating a sequential prediction chain where the mathematical outputs of the Development Occurrence hazard ($P(D)$) formally condition the expected project scale (E[units]), which in turn parameterizes a cumulative institutional friction pathway through each subsequent administrative horizon (H_1 to H_3), municipal planners could theoretically simulate end-to-end future neighborhood conflict trajectories on currently empty parcels years before an application is even conceived.

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A Research Project Overview

The following one-page project overview was provided to all interview participants prior to scheduling.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Daniel Hardesty Lewis
M.S. Candidate, Urban Planning

Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact
Daniel Hardesty Lewis
Graduate Researcher
Email: dl3645@columbia.edu

IRB Contact
Columbia University Morningside IRB
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

B IRB Protocol Summary

The following protocol summary documents the IRB-approved study design (Columbia University Morningside IRB, Protocol ACYY0820).

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation

Protocol Summary
ACYY0820

Study Protocol Summary

Predicting NIMBYism in Austin's Housing Reform Era

Key Personnel

Principal Investigator (PI): Hiba Bou Akar, Ph.D.
Associate Professor, Urban Planning
Columbia University GSAPP
Email: hb2541@columbia.edu

Student Researcher: Daniel Lewis
M.S. Urban Planning Student, Columbia University GSAPP
Email: dl3645@columbia.edu

IRB Contact

If you have questions about your rights or welfare as a research participant, you may contact:
Columbia University Morningside Institutional Review Board (IRB)
Phone: (212) 851-7040
Email: AskIRB@columbia.edu

Study Purpose

Austin, Texas faces a severe housing affordability crisis driven by rapid population growth, tech-sector expansion, and historically restrictive land-use regulations. Recent housing reforms have coincided with organized neighborhood opposition to development using tools such as protest petitions, public testimony, and legal challenges. This study aims to:

1. Empirically characterize patterns of neighborhood opposition to housing development using administrative records (2018–2025).
2. Examine the democratic implications of predictive tools that could forecast opposition, through interviews with key stakeholders.

Research Questions

Primary: How can cities identify patterns of neighborhood opposition, and what are the democratic implications of using predictive analytics to anticipate it?
Sub-questions:

- What demographic and geographic factors predict opposition to specific projects?
- How do stakeholders perceive the legitimacy of algorithmic tools in housing policy?
- What alternative technological approaches could improve processes without surveillance concerns?

Study Design

This mixed-methods, minimal-risk study combines:

- **Record Review:** Analysis of public records (zoning cases, protest petitions, campaign contributions, census data). De-identified datasets will use coded IDs.
- **Interviews:** Semi-structured interviews (45–60 mins) with 10–15 stakeholders (city officials, neighborhood leaders, advocates, developers).
- **Observation:** Review of public meetings (Council, commissions).

Participation Details

Interviews: Conducted in-person or via secure videoconference. With permission, they are audio-recorded for transcription. Participation is voluntary; you may decline to answer or stop at any time.
Risks: Minimal risk. Potential breach of confidentiality is mitigated by secure storage, encryption, and reporting only aggregate/de-identified results.
Benefits: No direct benefit. Indirect benefits include informing more equitable planning practices.

C Semi-Structured Interview Guide

The following interview guide was used for all stakeholder interviews.

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin's Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of "neighborhood opposition" to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

D Sample Protest Petition

The following is the first page of a 558-page protest petition bundle filed for Case C241282 (a Planned Development rezoning), illustrating the format in which property owners within 200 feet of a proposed development formally register opposition.

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			**Total Area of Petitioner:**	**Total %**
	Stacy Meeks		**84,008.88**	**28.67%**

E Opposition Model Precision-Recall Curves

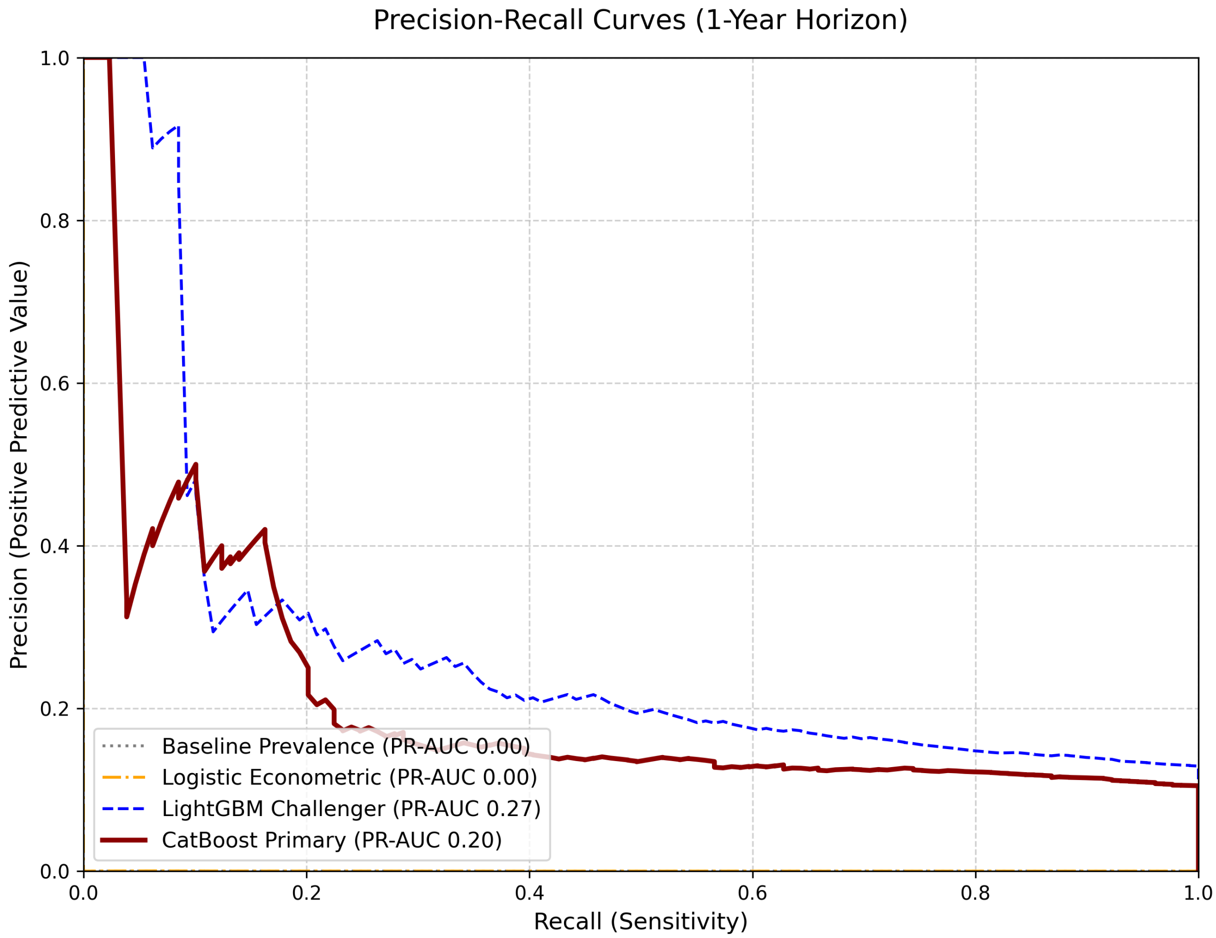

Figure 17: Organized opposition precision-recall curves by model architecture: CatBoost, Random Forest, ElasticNet Logistic, and V-REx.

F Temporal Panel Data Structure

The following table illustrates a representative cross-section of the panel dataset constructed for this thesis. Each parcel-year constitutes a unique observation, enriched with demographic variables (interpolated ACS), spatial protest indicators, and property tax valuations (TCAD EARS).

parcel_id	year	protest_buffer	owner_exempt	tract_wealth_gap	prop_med_rent	case_white_pct
0209060502	2008	True	True	12500	1600	0.58
0209060503	2008	True	True	11200	1600	0.58
0209060506	2008	True	True	14000	1650	0.58
0209060901	2008	True	False	13400	1450	0.58
1911420101	2008	False	True	-8000	1150	0.42
1911420102	2008	False	False	-6500	1150	0.42

G Methodological Appendix: Expanded Details

G.1 Variable Dictionary

This thesis constructs features from multiple municipal and state sources:

- **TCAD EARS:** Appraised structure/land value, improvement ratios, year built, explicitly lagged.
- **Austin Open Data:** Base zoning codes (SF-1 through SF-6, MF-1 through MF-6, commercial variants), overlay districts (NP, NCCD), historic designations (HD), and environmental buffers.
- **ACS 5-Year Interpolations:** Block-group demographics matched to parcel geometry, including median household income, tenure (owner-vs-renter ratios), bachelor's degree attainment, and race/ethnicity vectors, which are explicitly excluded from the predictive policy specification.

G.2 Hyperparameter Search Grids

Tree-based models (CatBoost, Random Forest) utilized the following randomized search bounds across 5-fold temporal cross validation:

- **learning_rate:** Log-uniform [0.01, 0.3]
- **depth:** [4, 6, 8, 10]
- **l2_leaf_reg:** Uniform [1, 10]
- **auto_class_weights:** [None, Balanced, SqrtBalanced]

Selected CatBoost models uniformly converged on Depth=6, Learning Rate $\approx$ 0.05, and 'Balanced' class weighting.

G.3 Subgroup Definitions & Fairness Testing

To calculate the False Negative Rate (FNR) gaps:

- **Geography:** Subgroups defined by the 10 Austin City Council Districts.
- **Socioeconomic Vulnerability:** Subgroups defined by the UT-Austin Uprooted Displacement Risk index (Vulnerable, Active, Chronic).[4]

G.4 Natural Language Processing Architecture (Exploratory Text Features)

Text signals from public hearings post-date the initial filing date prediction horizon and are therefore explored only at later administrative stages. Because these features are unavailable at the time of filing, they are treated as exploratory supplements rather than inputs to the primary opposition model.

The pipeline proceeds in three stages:

[4] Categories follow the 2024 City of Austin update. The original 2018 Uprooted report used a six-category gentrification typology (Susceptible, Early Type 1, Early Type 2, Dynamic, Late, Continued Loss) based on Bates (2013).

1. **Transcription.** Audio recordings of City Council and Planning Commission hearings are transcribed locally using Whisper (faster-whisper). Each transcript is linked to its originating zoning case number, producing a corpus of 1,003 case-level text documents.

2. **Seed labeling via large language model.** A randomized sample of 50 transcripts is classified by a locally hosted language model (Llama 3.2, 3B parameters). For each transcript, the model is prompted to return two outputs:

   - **Stance:** Whether the overall testimony is supportive, opposed, or neutral toward the proposed zoning change.
   - **Argument frames:** Which of five predefined thematic categories appear in the testimony: (1) traffic and congestion, (2) infrastructure capacity, (3) displacement and affordability, (4) neighborhood character and compatibility, and (5) procedural fairness. These categories were selected based on recurring themes identified in the urban planning literature on participatory opposition.

   The language model responses are parsed into structured binary labels (frame present or absent) and a three-class stance label. These 50 labeled cases serve as the training seed.

3. **Supervised expansion across the full corpus.** Using the 50 seed labels as training data, six logistic regression classifiers are trained—one for stance and one for each of the five argument frames—with TF-IDF unigram features (1,000-term vocabulary, English stop words removed) as inputs. Each classifier then predicts a continuous probability for every case in the full 1,003-transcript corpus. The result is a case-level matrix of six predicted probabilities: the likelihood that the hearing testimony is oppositional, and the likelihood that each argument frame is present.

This approach is intentionally lightweight. The seed sample is small (50 cases), and the classifiers use simple linear models over bag-of-words features. Consequently, the predicted probabilities should be interpreted as coarse thematic indicators rather than precise measurements. The goal is to test whether even rough text signals carry discriminative power for opposition prediction at later horizons, not to build a production-grade NLP system.

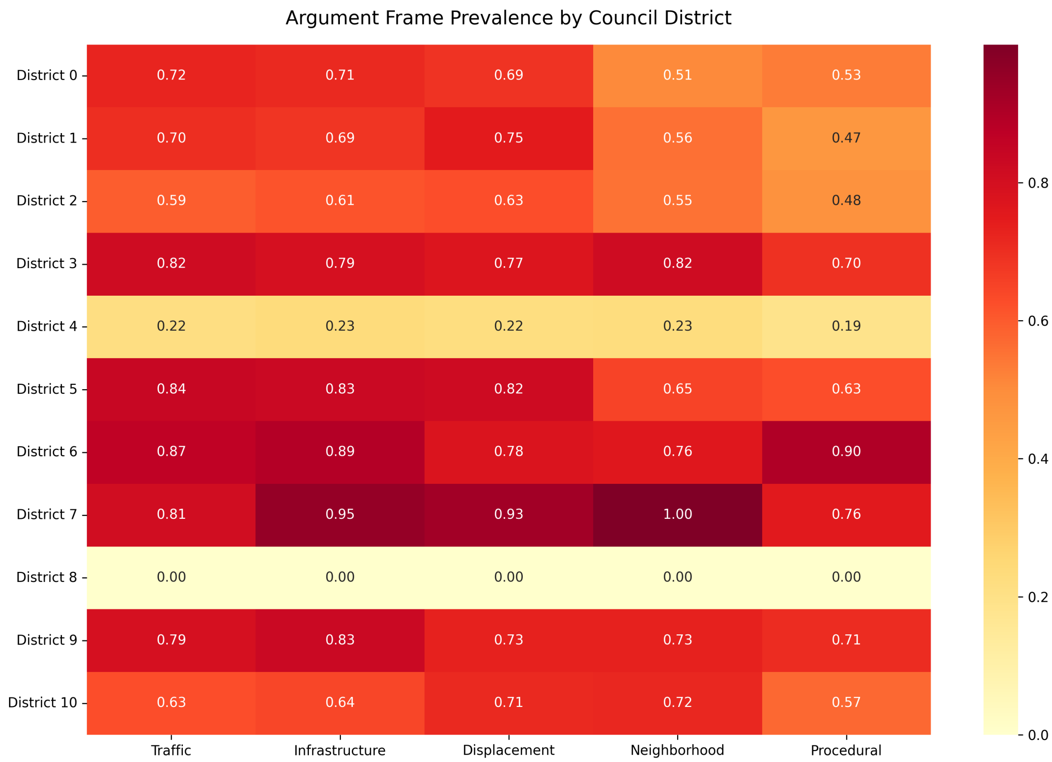

Figure 18: Mean predicted probability of each argument frame by Council District. Each of the 10 Austin Council Districts is shown as a row; columns represent the five argument frames classified from hearing transcripts. Darker cells indicate districts where a given frame appears more frequently in hearing testimony. Values are normalized to the maximum cell value for visual comparison.

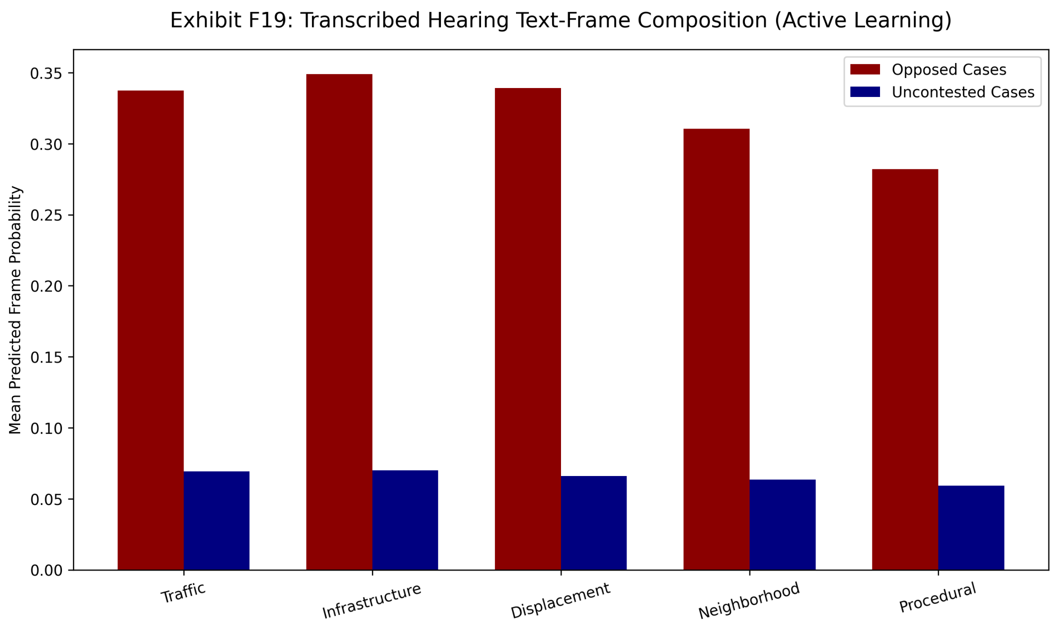

Figure 19: Mean predicted argument-frame probability for cases with a valid protest petition filed (Opposed) versus cases without (Uncontested). Bars compare how frequently each thematic frame appears in hearing testimony across the two groups.